Concept Note: AI-Driven Logistics Optimization for Carbon Reduction

1) Project Name

EcoLogix AI: Reducing Carbon Footprint in E-Commerce Logistics through Predictive Analytics.

2) Overview

This project aims to minimize the environmental impact of e-commerce logistics operations. By analyzing delivery routes, vehicle types, and package weights within the dataset, we intend to develop AI-based strategies to optimize carbon emissions. The project specifically focuses on enhancing sustainability in the logistics sector by reducing the utilization of high-emission vehicles.

3) Background & SDG Alignment

The logistics sector is responsible for a significant portion of global carbon emissions. While traditional methods often focus on route efficiency, they may treat environmental sustainability as a secondary priority.

- SDG 13 (Climate Action): Utilizing data analytics to lower emissions derived from logistics operations.
- SDG 9 (Industry, Innovation, and Infrastructure): Establishing more efficient and eco-friendly transportation networks.
- SDG 11 (Sustainable Cities and Communities): Reducing air pollution in urban delivery scenarios (Urban Last Mile).

4) Key Objectives & Research Questions

a) Research Questions

The project seeks to address the following key research questions:

- Emission Factors: To what extent do vehicle type and package weight determine the total carbon emission (kgCO2e)?
- Distance Efficiency: How does the correlation between distance traveled and emission increase vary across different operational routes (e.g., Urban Last Mile vs. Inter-City)?
- AI Potential: How accurately can the carbon footprint of the next delivery be predicted using existing historical logistics data?
- Sustainability Impact: What percentage of total emission reduction can be achieved by scaling "Eco-friendly" logistics options?

b) Key Steps

The roadmap for the successful completion of the project is as follows:

1. Data Collection & Preparation: Acquiring the open-source e-commerce logistics dataset and importing it into the Python environment for analysis
2. Exploratory Data Analysis (EDA): Checking for missing values, extracting descriptive statistics, and visualizing fundamental trends.
3. Feature Analysis: Identifying the parameters with the highest impact on carbon emissions (Feature Importance) using statistical methods.
4. Predictive Model Design: Constructing an AI model framework based on cleaned data to optimize or predict carbon emissions.
5. Reporting & Presentation: Linking analysis insights with SDG targets to finalize the Concept Note and presentation materials.

```
Python 3.11.10 | packaged by conda-forge | (main, Oct 16 2024, 01:17:14) [MS
v.1941 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 8.29.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile 'C:/Users/Elif Elvin Acar/OneDrive/Desktop/Yeni klasör/
untitled8.py' --wdir
--- Veri Seti Özeti ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12000 entries, 0 to 11999
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Transaction_ID          12000 non-null  object
 1   Date                    12000 non-null  object
 2   Origin_Facility         12000 non-null  object
 3   Destination_City        12000 non-null  object
 4   Vehicle_Type            12000 non-null  object
 5   Route_Type              12000 non-null  object
 6   Distance_KM             12000 non-null  float64
 7   Package_Weight_KG       12000 non-null  float64
 8   Traffic_Conditions      12000 non-null  object
 9   Carbon_Emission_kgCO2e  12000 non-null  float64
```

5) Exploratory Data Analysis (EDA) Insights

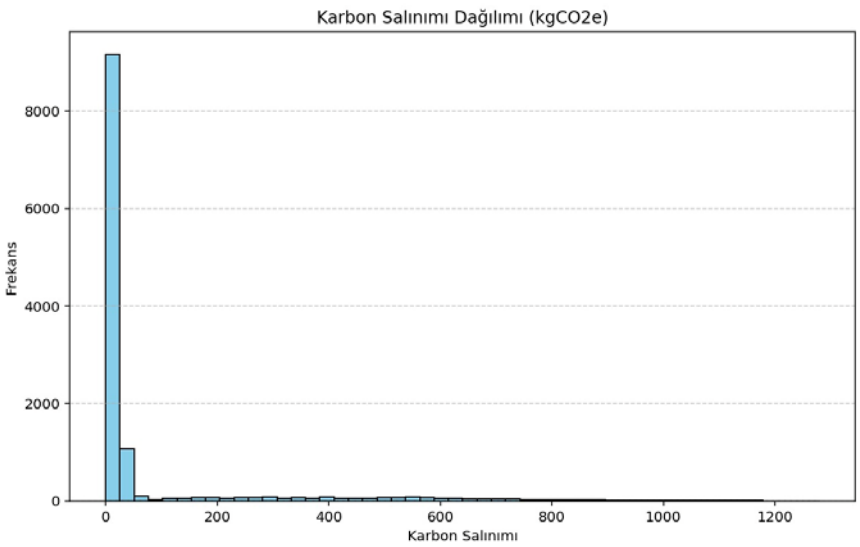
The following key findings were derived from the analysis:

- **Dataset Summary:** A total of 12,000 transactions were analyzed, with no missing (null) values detected.

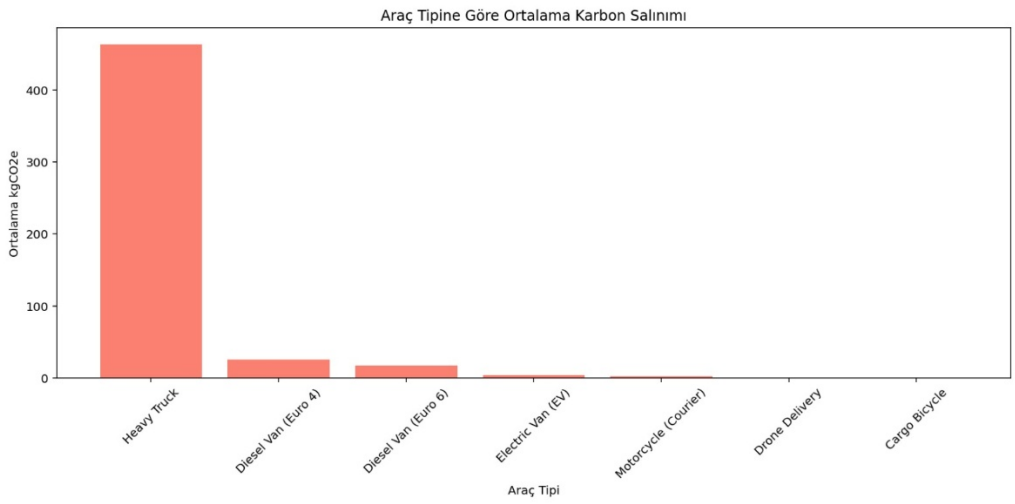
```
25%       11.600000           8.800000                0.093000
0.000000
50%       37.500000          17.300000                4.150000
0.000000
75%      114.600000          26.100000               24.091250
1.000000
max      799.900000        4985.500000             1280.173000
1.000000

--- Eksik Değer Kontrolü ---
Transaction_ID            0
Date                      0
Origin_Facility           0
Destination_City          0
Vehicle_Type              0
Route_Type                0
Distance_KM               0
Package_Weight_KG         0
Traffic_Conditions        0
Carbon_Emission_kgCO2e    0
Is_Eco_Friendly           0
dtype: int64
```

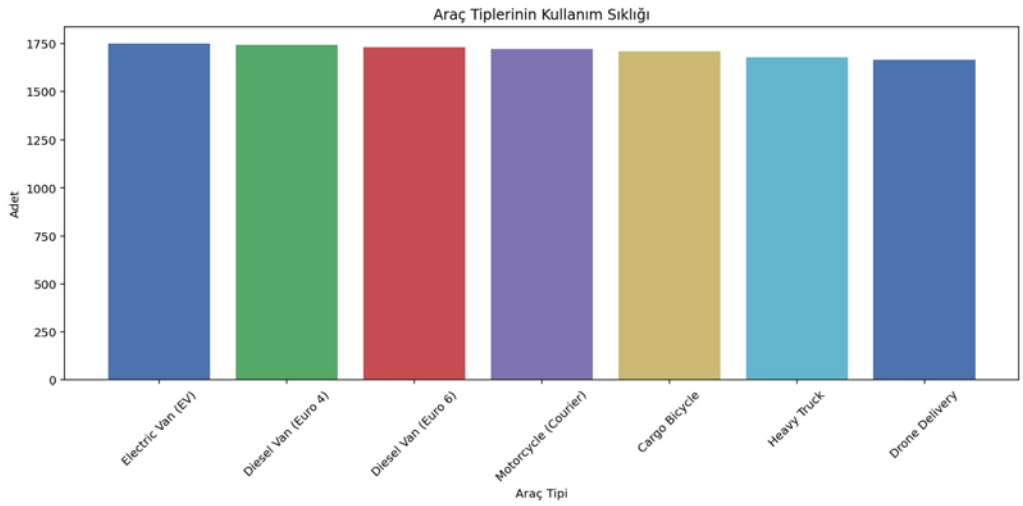
```
 10  Is_Eco_Friendly          12000 non-null  int64
dtypes: float64(3), int64(1), object(7)
memory usage: 1.0+ MB
None

--- Betimsel İstatistikler ---
        Distance_KM  Package_Weight_KG  Carbon_Emission_kgCO2e
Is_Eco_Friendly
count  12000.000000       12000.000000            12000.000000
12000.000000
mean     102.851408         396.355458               71.994260
0.427000
std      163.538763        1064.274964              180.524361
0.494663
min        1.000000           0.500000                0.000000
0.000000
25%       11.600000           8.800000                0.093000
0.000000
50%       37.500000          17.300000                4.150000
0.000000
75%      114.600000          26.100000               24.091250
1.000000
max      799.900000        4985.500000             1280.173000
1.000000
```

- **Carbon Emission Distribution:** While emissions are largely concentrated at lower levels (0-100 kgCO2e), outliers reaching up to 1,280 kgCO2e were observed in certain operations.

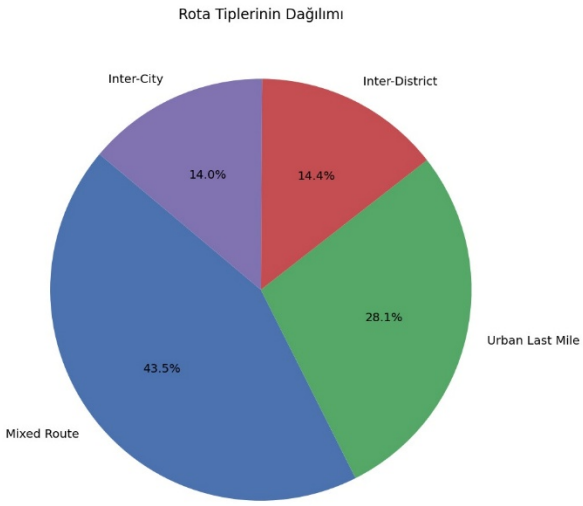


- **Vehicle-Based Analysis:** "Heavy Trucks" exhibit significantly higher average carbon emissions compared to all other vehicle types. Conversely, Electric Vans and Drone deliveries produce minimal emissions.

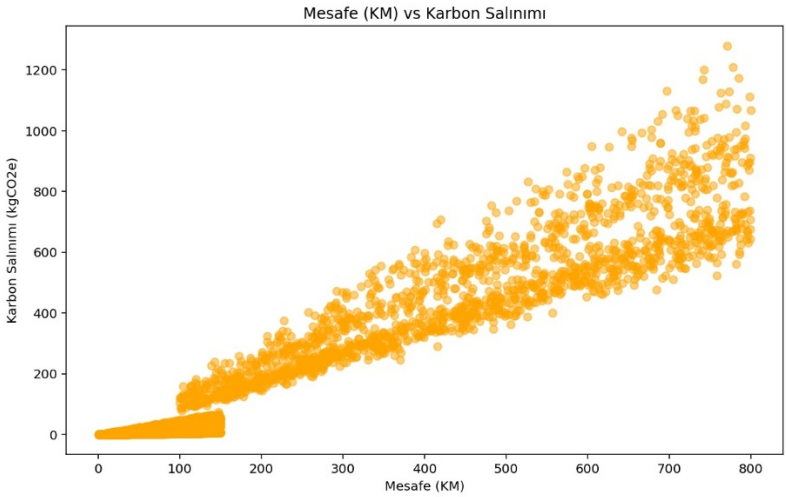




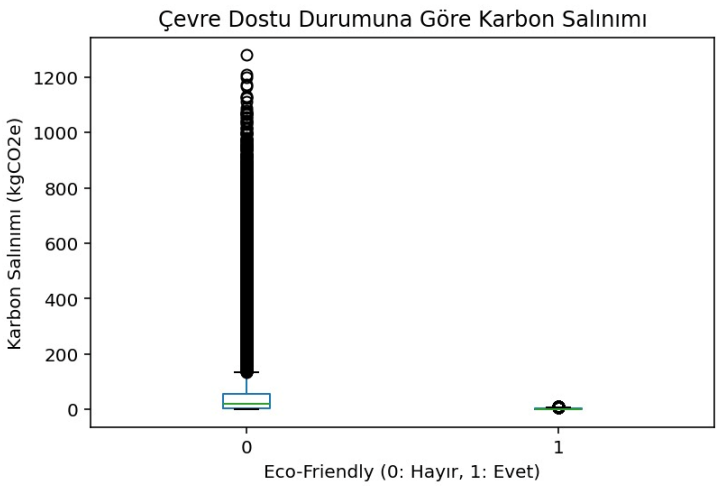
- **Route Analysis:** 43.5% of shipments occur on "Mixed Routes," while 28.1% are "Urban Last Mile" deliveries.



- **Correlation:** A clear linear increase in carbon emissions is observed as the distance (Distance_KM) increases.



- **Eco-Friendly Status:** Transactions flagged as "Is_Eco_Friendly" show a significantly lower emission profile compared to non-flagged transactions.



6) Methodology & Tools

- Data Preprocessing: Performed using the Python (Pandas) library for data cleaning and descriptive statistical calculations.
- Visualization: Various charts, including histograms, bar charts, pie charts, and boxplots, were generated using the Matplotlib library.
- Modeling Approach: In the next phase, algorithms such as Ridge Regression or Echo State Networks (ESN) will be evaluated to predict and optimize carbon emissions.

7) Appendix

Appendix A: Technical Documentation and Data Analysis Script

The following figure presents the Python source code developed to execute the Exploratory Data Analysis (EDA) and visualization phases of the EcoLogix AI project. The script utilizes the Pandas library for structured data manipulation and the Matplotlib library for generating the statistical distributions and correlations discussed in Section 5.

```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('ecommerce_logistics_carbon_emissions_v1.csv')

# 1. Temel Bilgiler ve İstatistikler
print("--- Veri Seti Özeti ---")
print(df.info())
print("\n--- Betimsel İstatistikler ---")
print(df.describe())
print("\n--- Eksik Değer Kontrolü ---")
print(df.isnull().sum())

# Renk paleti tanımlayalım
colors = ['#4c72b0', '#55a868', '#c44e52', '#8172b2', '#ccb974', '#64b5cd']

# 2. Karbon Salınımı Dağılımı (Histogram)
plt.figure(figsize=(10, 6))
plt.hist(df['Carbon_Emission_kgCO2e'], bins=50, color='skyblue', edgecolor='black')
plt.title('Karbon Salınımı Dağılımı (kgCO2e)')
plt.xlabel('Karbon Salınımı')
plt.ylabel('Frekans')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

# 3. Araç Tiplerinin Kullanım Sıklığı (Bar Chart)
plt.figure(figsize=(12, 6))
vehicle_counts = df['Vehicle_Type'].value_counts()
plt.bar(vehicle_counts.index, vehicle_counts.values, color=colors[:len(vehicle_counts)])
plt.title('Araç Tiplerinin Kullanım Sıklığı')
plt.xlabel('Araç Tipi')
plt.ylabel('Adet')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 4. Mesafe ve Karbon Salınımı İlişkisi (Scatter Plot)
plt.figure(figsize=(10, 6))
plt.scatter(df['Distance_KM'], df['Carbon_Emission_kgCO2e'], alpha=0.5, color='orange')
plt.title('Mesafe (KM) vs Karbon Salınımı')
plt.xlabel('Mesafe (KM)')
plt.ylabel('Karbon Salınımı (kgCO2e)')
plt.show()

# 5. Araç Tipine Göre Ortalama Karbon Salınımı (Bar Chart)
plt.figure(figsize=(12, 6))
avg_carbon_vehicle = df.groupby('Vehicle_Type')['Carbon_Emission_kgCO2e'].mean().sort_values(ascending=False)
plt.bar(avg_carbon_vehicle.index, avg_carbon_vehicle.values, color='salmon')
plt.title('Araç Tipine Göre Ortalama Karbon Salınımı')
plt.xlabel('Araç Tipi')
plt.ylabel('Ortalama kgCO2e')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 6. Rota Tiplerinin Dağılımı (Pie Chart)
plt.figure(figsize=(8, 8))
route_counts = df['Route_Type'].value_counts()
plt.pie(route_counts.values, labels=route_counts.index, autopct='%1.1f%%', startangle=140, colors=colors)
plt.title('Rota Tiplerinin Dağılımı')
plt.show()

# 7. Çevre Dostu Durumuna Göre Karbon Salınımı (Boxplot)
plt.figure(figsize=(8, 6))
df.boxplot(column='Carbon_Emission_kgCO2e', by='Is_Eco_Friendly', grid=False)
plt.title('Çevre Dostu Durumuna Göre Karbon Salınımı')
plt.suptitle('') # Otomatik başlığı kaldır
plt.xlabel('Eco-Friendly (0: Hayır, 1: Evet)')
plt.ylabel('Karbon Salınımı (kgCO2e)')
plt.show()
```